

Two-factor ANOVA under balanced design without replicate

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LSM

$$Y_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ij} \quad \begin{matrix} i=1, \dots, a \\ j=1, \dots, b \end{matrix}$$

Ex. An analyst studied the premium for auto insurance. Six cities were selected to represent different sizes (small, medium, large) and different regions with the state (north, south). There is exactly one city for each combination of size and region.

Data

		Region (factor B)		
		North	South	
Size (factor A)	small	140	100	$\bar{Y}_{1.} = 120$
	medium	210	180	$\bar{Y}_{2.} = 195$
	large	220	200	$\bar{Y}_{3.} = 210$
		$\bar{Y}_{.1} = 190$	$\bar{Y}_{.2} = 160$	$\bar{Y}_{..} = 175$

$$Y_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ij} \quad \begin{matrix} i=1, \dots, a \\ j=1, \dots, b \end{matrix}$$

$$\sum_i \alpha_i = 0 \quad \sum_j \beta_j = 0 \quad \sum_i \gamma_{ij} = 0 \quad \sum_j \gamma_{ij} = 0$$

Step 1: Get an idea about the presence or absence of interaction by fitting an additive model (i.e., without interaction) to the data, and estimating the treatment means

$$\mu_{ij} = \mu + \alpha_i + \beta_j = \mu + \mu_{i.} - \mu + \mu_{.j} - \mu$$

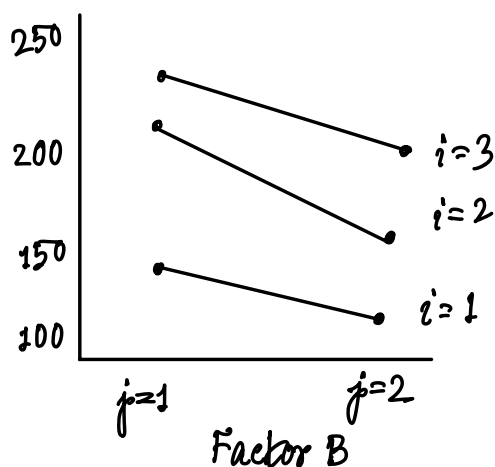
Let under additive model

$$\hat{\mu}_{ij} = \bar{Y}_{i.} + \bar{Y}_{.j} - \bar{Y}_{..}$$

Draw an interaction plot based on the estimated ~~treatment means~~ observations
 $\hat{Y}_{ij}$ = new estimates of μ_{ij}

	North	South	
small	$\hat{\mu}_{11} = 135$	$\hat{\mu}_{12} = 105$	$\bar{Y}_{1.} = 120$
medium	$\hat{\mu}_{21} = 210$	$\hat{\mu}_{22} = 180$	$\bar{Y}_{2.} = 195$
large	$\hat{\mu}_{31} = 225$	$\hat{\mu}_{32} = 195$	$\bar{Y}_{3.} = 210$
	$\bar{Y}_{.1} = 190$	$\bar{Y}_{.2} = 160$	$\bar{Y}_{..} = 175$

Model



Step 2:
ANOVA decomposition

$$SSTO = SSA + SSB + SSAB$$

(since $SSE = 0$)

$$Y_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$$

If there is no interaction then SSAB acts as SSE under the corresponding additive model.

$$SSTO = \sum_i \sum_j (Y_{ij} - \bar{Y}_{..})^2$$

$$SSAB = \sum_i \sum_j (Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})^2$$

$$= \sum_i \sum_j (Y_{ij} - \hat{\mu}_{ij})^2$$

est of μ_{ij} under additive model

$$df(SSAB) = (a-1)(b-1)$$

Then we could use $MSAB = \frac{SSAB}{(a-1)(b-1)}$ as an estimator of σ^2 .

Problem with testing for no interaction

If we try to estimate all v_{ij} 's (with no structural constraints) then there is no degree of freedom left to estimate the variance σ^2

A more limited objective: specify certain kinds of interactions that may be easy to interpret and check for their presence.

Tukey's proposal: Check for the presence of multiplicative interaction.

$$\gamma_{ij} = \theta \alpha_i \beta_j \text{ for some scalar } \theta$$

(Note. $\sum_i \gamma_{ij} = 0$ for all j
 $\sum_j \gamma_{ij} = 0$ for all i
 $\leftrightarrow \theta = 0$)

Under H_0 (no interaction) : $Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$

Under H_0 (no interaction): $Y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij}$
 Under H_A (multiplicative interaction): $Y_{ij} = \mu + \alpha_i + \beta_j + \theta \alpha_i \beta_j + \epsilon_{ij}$

Estimating θ : Use one step iterated least squares procedure

i.e. est $\{\alpha_i\}, \{\beta_j\}, \mu$ from (red) to obtain $\hat{\mu}, \hat{\alpha}_i, \hat{\beta}_j$
and est θ from $\min_{\theta} \sum_i \sum_j (y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \theta \hat{\alpha}_i \hat{\beta}_j)^2$

Note: If we define

$$\tilde{Y}_{ij} = \hat{\mu} + \hat{\alpha}_i + \hat{\beta}_j = \bar{Y}_{i.} + \bar{Y}_{.j} - \bar{Y}_{..}$$

and $\tilde{\epsilon}_{ij} = Y_{ij} - \tilde{Y}_{ij}$ then

$$\hat{\theta} = \frac{\sum_i \sum_j \tilde{\epsilon}_{ij} \hat{\alpha}_i \hat{\beta}_j}{\sum_i \sum_j (\hat{\alpha}_i \hat{\beta}_j)^2} = \frac{\sum_i \sum_j \tilde{\epsilon}_{ij} \hat{\alpha}_i \hat{\beta}_j}{\sum_i \hat{\alpha}_i^2 \sum_j \hat{\beta}_j^2}$$

Notice that $\{\tilde{Y}_{ij}\}$ and $\{\tilde{\epsilon}_{ij}\}$ are independent

Using this, we can show that (Exc.)

$$E(\hat{\theta} | \tilde{Y}) = \frac{\theta \left(\sum_i \alpha_i \hat{\alpha}_i \right) \left(\sum_j \beta_j \hat{\beta}_j \right)}{\left(\sum_i \alpha_i^2 \right) \left(\sum_j \beta_j^2 \right)}$$

$$\text{Var}(\hat{\theta} | \tilde{Y}) = \frac{\sigma^2}{\left(\sum_i \alpha_i^2 \right) \left(\sum_j \beta_j^2 \right)}$$

$SSE_{\text{red}} = SSAB = (SSE \text{ under additive model})$

$$SSE_{\text{full}} = \sum_i \sum_j (Y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\theta} \hat{\alpha}_i \hat{\beta}_j)^2$$

mult.
interaction

$$= \sum_i \sum_j \underbrace{(\tilde{\epsilon}_{ij} - \hat{\theta} \hat{\alpha}_i \hat{\beta}_j)^2}_{\tilde{\epsilon}_{ij}^2}$$

Since $\tilde{\epsilon}_{ij} = \epsilon_{ij} + \hat{\theta} \hat{\alpha}_i \hat{\beta}_j$ and $\sum_i \hat{\alpha}_i = \sum_j \hat{\beta}_j = 0$,

We have $\sum_i \sum_j \tilde{\epsilon}_{ij}^2 = \sum_i \sum_j \hat{\epsilon}_{ij}^2 + \hat{\theta}^2 \left(\sum_i \alpha_i^2 \right) \left(\sum_j \beta_j^2 \right)$

$\parallel$ $\parallel$ $\parallel$
 SSE_{red} SSE_{full} $SSAB^*$ — (*)

(check $\sum_i \sum_j \epsilon_{ij} \hat{\theta} \hat{\alpha}_i \hat{\beta}_j = \hat{\theta} \left(\sum_i \sum_j \epsilon_{ij} \hat{\alpha}_i \hat{\beta}_j \right)$

$$= \hat{\theta} \cdot \hat{\theta} \left(\sum_i \hat{\alpha}_i^2 \right) \left(\sum_j \hat{\beta}_j^2 \right)$$

Using the fact that

- ① $\tilde{\epsilon}$ and $\tilde{Y}$ are indep
 - ② $\hat{\theta}$ is a linear function of $\tilde{\epsilon}$ cond. on $\tilde{Y}$
- and the decomposition (*) we claim that [conditionally on $\tilde{Y}$]

(a) $\frac{SSAB^*}{\sigma^2} \sim \chi^2_1(\delta_1^2)$ where $\delta_1^2 = \frac{\theta^2 M^2}{2\sigma^2}$

(b) $\frac{SSAB}{\sigma^2} \sim \chi^2_{(a-1)(b-1)}(\delta^2)$ $\delta^2 = \frac{\theta^2 \left(\sum_i \hat{\alpha}_i^2 \right) \left(\sum_j \hat{\beta}_j^2 \right)}{2\sigma^2}$

Difference $SSAB - SSAB^* = SSE_{full} = SSRem^*$

(c) $\frac{SSRem^*}{\sigma^2} \sim \chi^2_{ab-a-b}(\delta^2 - \delta_1^2)$

(Use Fisher-Cochran theorem cond. on $\tilde{Y}$)

$SSRem^*$ and $SSAB^*$ are conditionally independent given $\tilde{Y}$

ANOVA Table

Source	df	SS	MS
Factor A	$a-1$	SSA	$SSA/a-1$
Factor B	$b-1$	SSB	$SSB/b-1$
mult. interaction	1	$SSAB^*$	$SSAB^*$
Error	$ab-a-b$	$SSRem^*$	$SSRem^*/ab-a-b$
Total	$ab-1$		

Conditionally on $\tilde{Y}$, the distⁿ of

$$F_{AB}^* = \frac{SSAB^*/1}{SSRem^*/ab-a-b} \sim F_{1, ab-a-b}$$

← unconditional